

Discourse Representation Theory (DRT)

Week 9

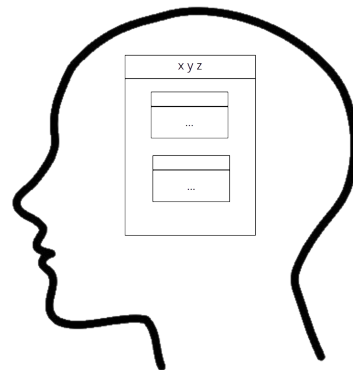
Slides and materials based on the courses by
Noortje Venhuizen and Mareike Hartmann

Recap: dynamic semantics

- Basic semantic value: context-change potential
- Existential quantification over: the discourse
- Quantification is: unselective

Discourse Representation Theory

- **Mentalist, representationalist**, and *dynamic* theory of the interpretation of discourse
- Ingredients:
 - Discourse Representation Structures (DRSs)
 - Construction procedure for DRSs
 - Model-theoretic interpretation (at the discourse level)



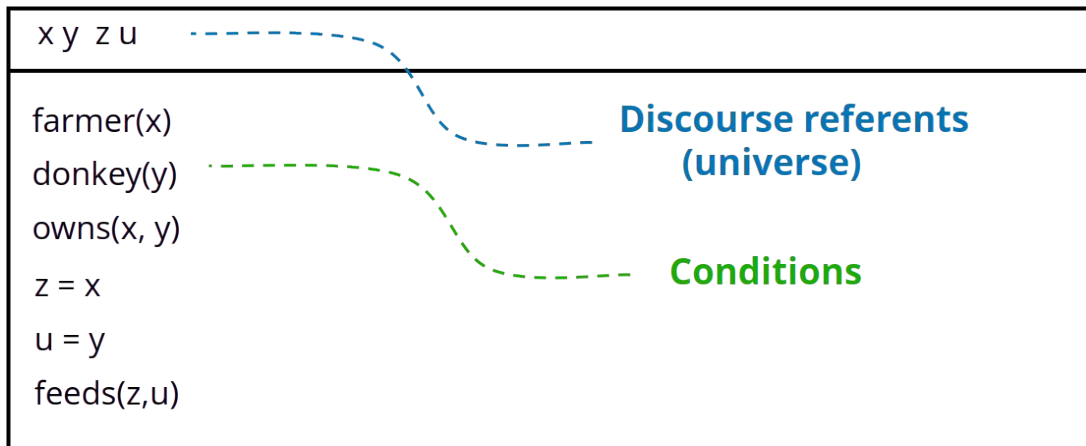
Basic features of DRT

- DRT models linguistic meaning as **anaphoric potential** (through DRS construction) plus truth conditions (through model embedding)
- DRT explains the ambivalent character of indefinite noun phrases:
 - Indefinite NPs are expressions that introduce new reference objects into the context (like DPL)

Discourse Representation Structures

- A context is represented as a **Discourse Representation Structure (DRS)** consisting of a set of **discourse referents** and a set of **conditions**

“A farmer owns a donkey. He feeds it.”

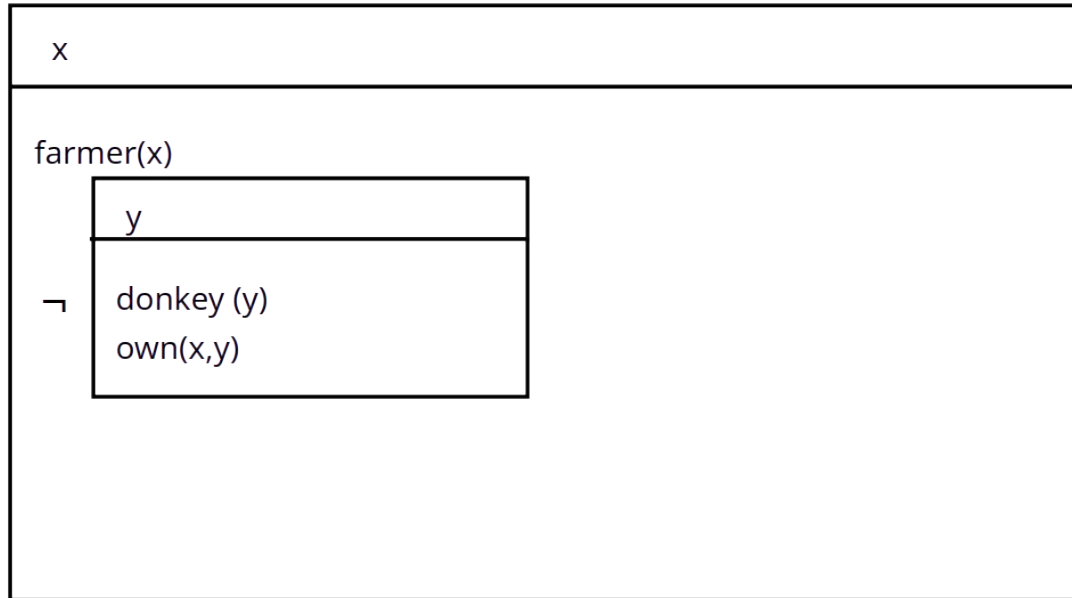


DRS syntax

- A discourse representation structure (DRS) K is a pair (U_K, C_K) , where:
 - $U_K \subseteq U_D$ and U_D is a set of **discourse referents**
 - C_K is a set of well-formed **DRS conditions**
- Well-formed DRS conditions:
 - $R(u_1, \dots, u_n)$ where: R is an n -place relation, $u_i \in U_D$
 - $u = v$ $u, v \in U_D$
 - $u = a$ $u \in U_D$, a is a constant
 - $\neg K_1$ K_1 is a DRS
 - $K_1 \Rightarrow K_2$ K_1 and K_2 are DRSs
 - $K_1 \vee K_2$ K_1 and K_2 are DRSs

DRS example

“A farmer does not own a donkey.”

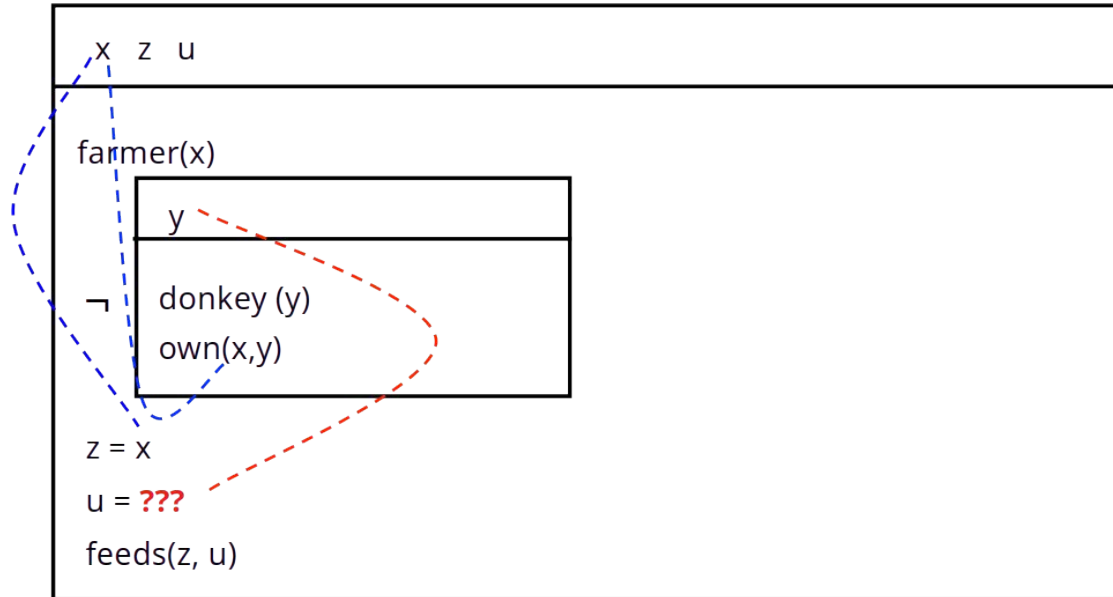


Non-accessible discourse referents

- “If a professor owns a book, he reads it. *It* has 300 pages.”
 - “It is not the case that a professor owns a book. *He* reads *it*.”
 - “Every professor owns a book. *He* reads *it*.”
 - “If every professor owns a book, *he* reads *it*.”
 - “Peter owns a book, or Mary reads *it*.”
 - “Peter reads a book, or Mary reads a newspaper article. *It* is interesting.”
-
- To explain this pattern, we need to formalize accessibility of discourse referents!

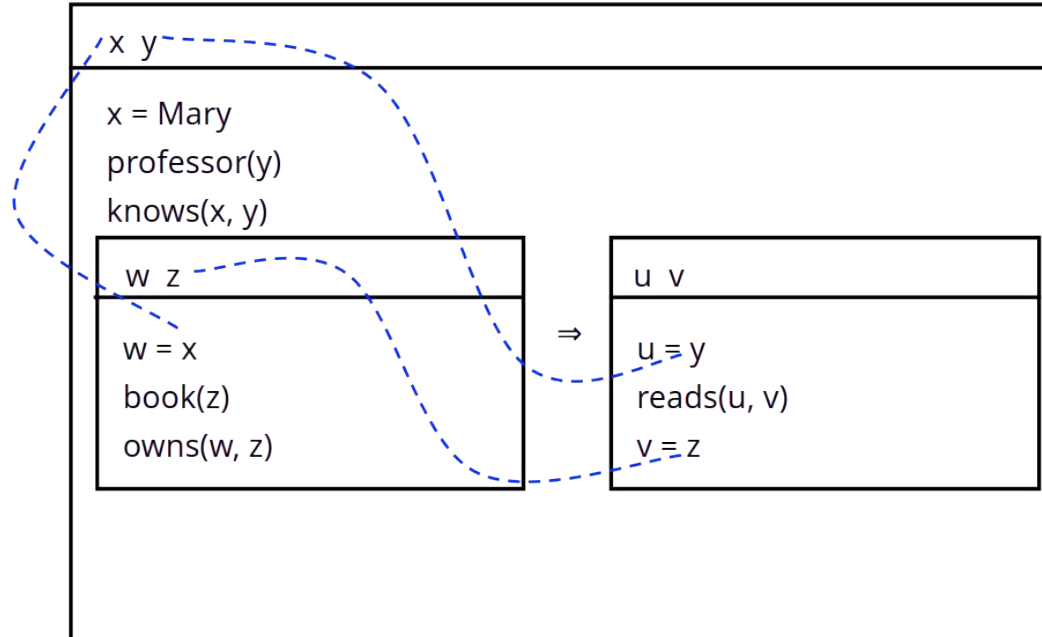
Anaphora and accessibility

“A farmer does not own a donkey. # He feeds it.”



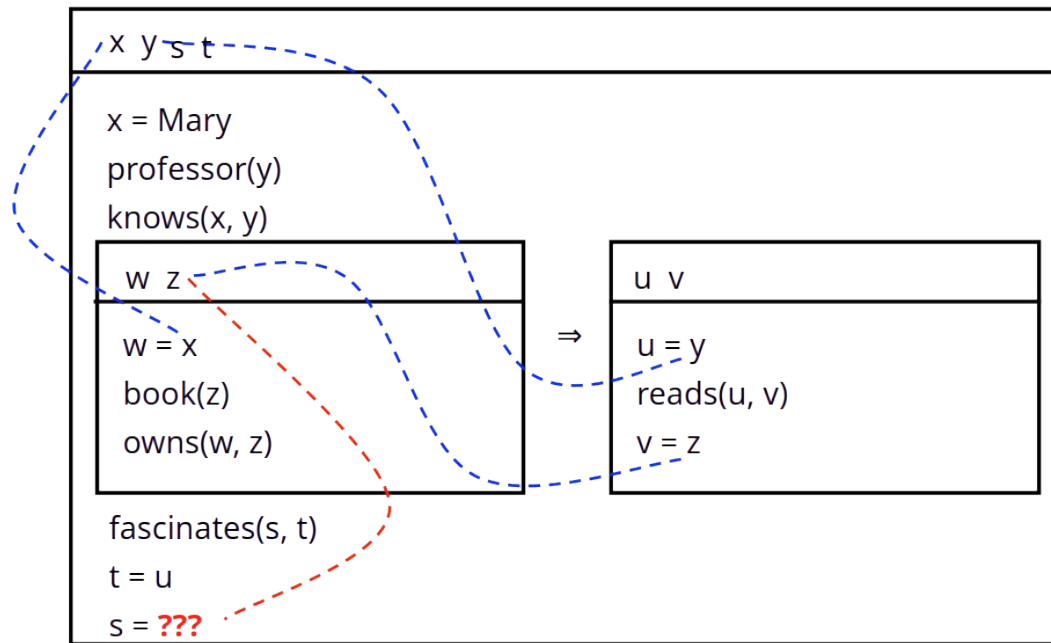
Anaphora and accessibility

“Mary knows a professor. If she owns a book, he reads it.”



Anaphora and accessibility

“Mary knows a professor. If she owns a book, he reads it. # It fascinates him.”



Accessible discourse referents

- The following discourse referents are accessible from a DRS condition:
 - Referents in the same local DRS
 - Referents in a superordinate DRS
 - Referents in an antecedent DRS, if the condition occurs in the consequent DRS
- We need a formal notion of DRS subordination

Subordination

- DRS K_1 is an **immediate sub-DRS** of a DRS $K = (U_K, C_K)$ iff
 - C_K contains a condition of the form: $\neg K_1$, $K_1 \Rightarrow K_2$, $K_2 \Rightarrow K_1$, $K_1 \vee K_2$ or $K_2 \vee K_1$
- DRS K_1 is a **sub-DRS** of DRS K (notation: $K_1 \leq K$) iff
 - $K_1 = K$, or
 - K_1 is an immediate sub-DRS of K , or
 - there is a DRS K_2 such that $K_1 \leq K_2$ and K_2 is an immediate sub-DRS of K
- DRS K_1 is a **proper sub-DRS** of DRS K iff
 - $K_1 \leq K$ and $K_1 \neq K$

Accessible discourse referents: formal definition

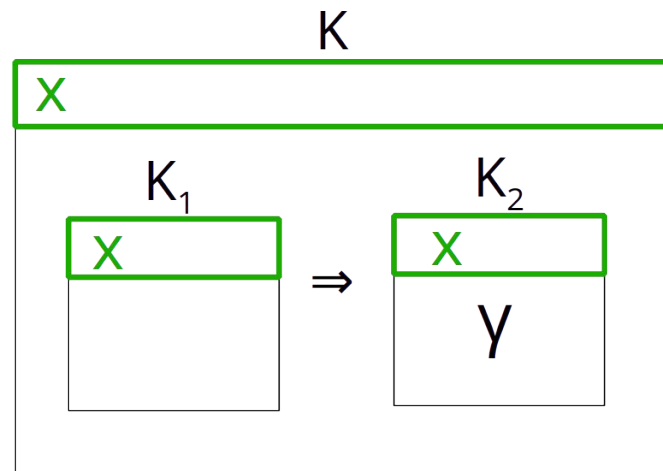
- Let K, K_n, K_m be DRSs such that:

$$K_n, K_m \leq K \text{ and } x \in U_{K_n} \text{ and } y \in C_{K_m}$$

- Then, **x is accessible from y in K** iff

- $K_m \leq K_n$ or
- there are $K_h, K_i \leq K$ such that:

$$K_n \Rightarrow K_h \in C_{K_i} \text{ and } K_m \leq K_h$$



Free and bound variables in DRT

- A DRS variable x , introduced in the conditions of DRS K_1 , is **bound** in global DRS K iff there exists a DRS $K_2 \leq K$, such that:
 - $x \in U_{K_2}$, and
 - K_2 is accessible from K_1 in K
- **Properness**: a DRS is proper iff it does not contain any free variables
- **Purity**: a DRS is pure iff it does not contain any **otiose declarations** of variables
 - i.e. $x \in U_{K_1}$ and $x \in U_{K_2}$ and $K_1 \leq K_2$

From text to DRS

Text

$\Sigma = \langle S_1, S_2, \dots, S_n \rangle$

Syntactic analysis

$\downarrow \quad \downarrow \quad \quad \downarrow$
 $P(S_1) \quad P(S_2) \quad \dots \quad P(S_n)$

DRS

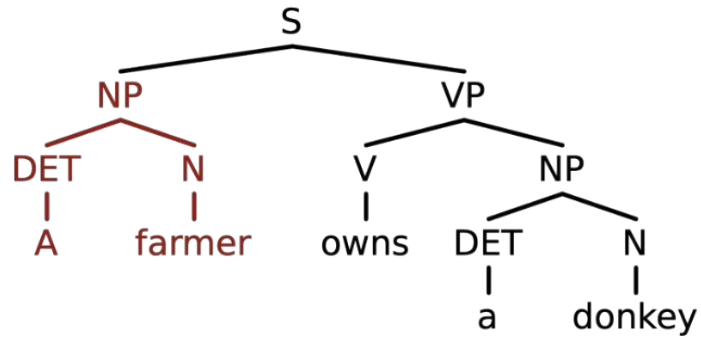
$\downarrow \quad \downarrow \quad \quad \downarrow$
 $K_1 \longrightarrow K_2 \longrightarrow \dots \longrightarrow K_n$

DRS construction algorithm

- Let the following be a well-formed, **reducible** DRS condition:
 - Conditions of form α or $\alpha(x_1, \dots, x_n)$, where α is a context-free parse tree
- DRS construction algorithm:
 - Given a text $\Sigma = (S_1, \dots, S_n)$, and a DRS $K_0 (= (\emptyset, \emptyset)$, by default)
 - Repeat for $i = 1, \dots, n$:
 - Add parse tree $P(S_i)$ to the conditions of K_{i-1}
 - Apply DRS construction rules to reducible conditions of K_{i-1} , until no more reduction steps are possible
 - The resulting DRS is K_i , the discourse representation of text Σ

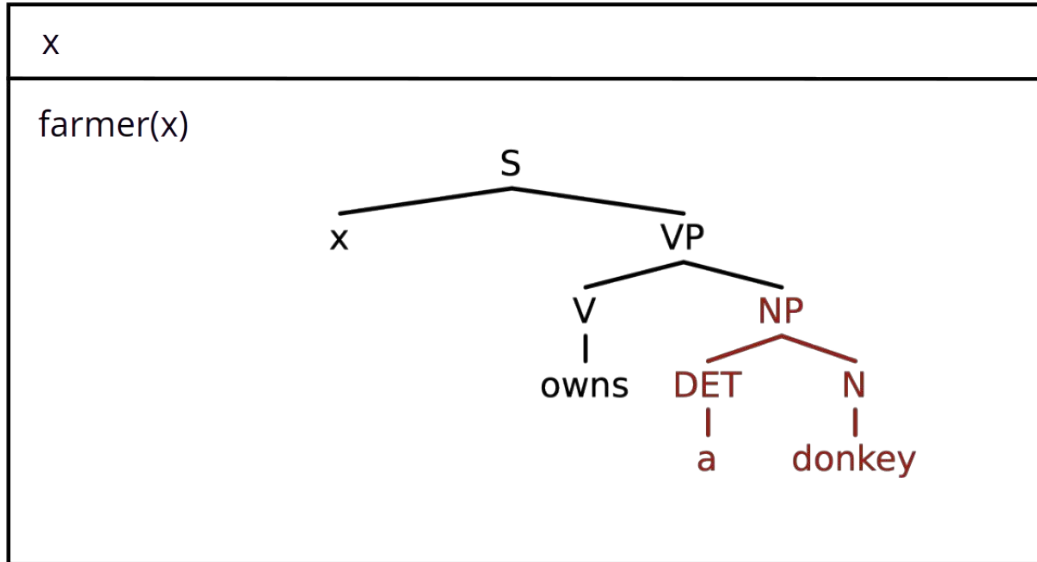
DRS construction: example

“A farmer owns a donkey. He beats it.”



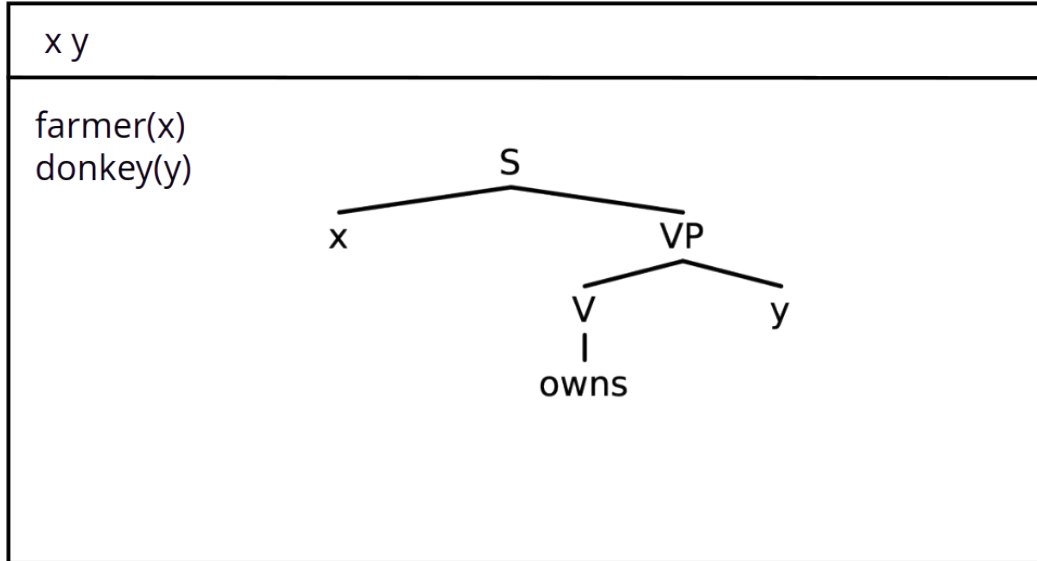
DRS construction: example

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DRS construction: example

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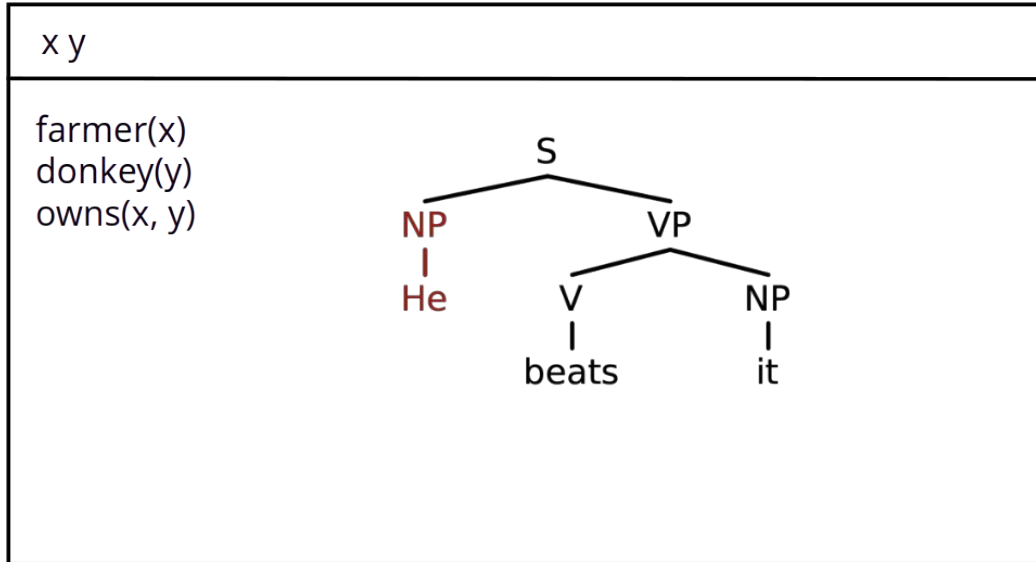
DRS construction: example

“A farmer owns a donkey. He beats it.”

x y
farmer(x) donkey(y) owns(x, y)

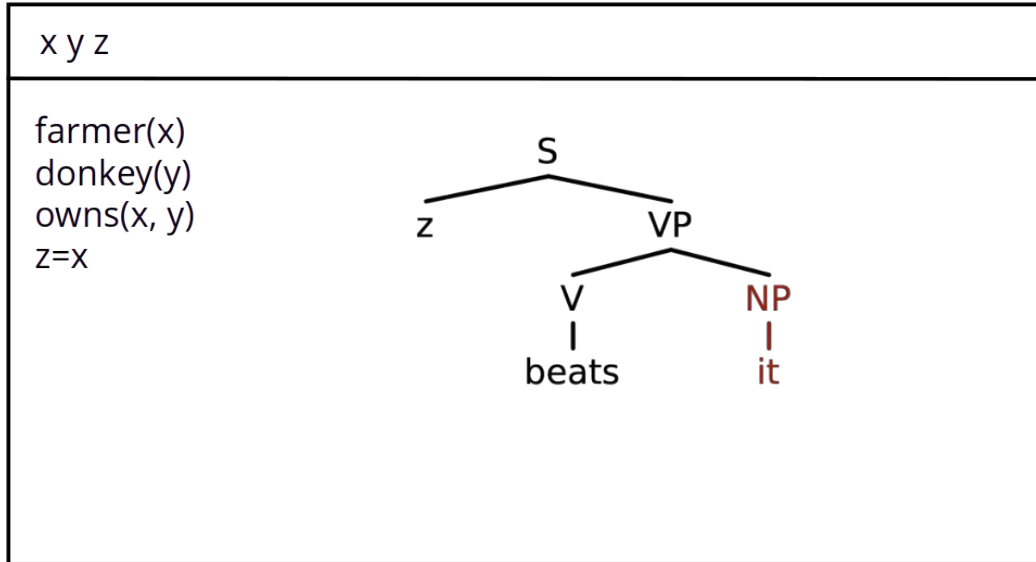
DRS construction: example

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DRS construction: example

“A farmer owns a donkey. He beats it.”



DRS construction: example

“A farmer owns a donkey. He beats it.”

x y z u
farmer(x) donkey(y) owns(x, y) z = x u = y beat(z, u)

Construction rules: indefinite NPs

- given a reducible condition α in DRS K , with $[S [NP \beta] [VP \gamma]]$ or $[VP [V \gamma] [NP \beta]]$ as a substructure, such that $\beta = \varepsilon\bar{\delta}$, where ε is an indefinite article:
 - (i) add a new discourse referent x to U_K
 - (ii) replace β in α by x
 - (iii) add $\bar{\delta}(x)$ to C_K

Construction rules: proper names

- Given a global DRS K^* , and some $K \leq K^*$, such that α is a reducible condition in DRS K , with $[S [NP \beta] [VP \gamma]]$ or $[VP [V \gamma] [NP \beta]]$ as a substructure, such that β is a proper name
 - (i) add a new discourse referent x to U_{K^*}
 - (ii) replace β in α by x
 - (iii) add $x = \beta$ to C_{K^*}

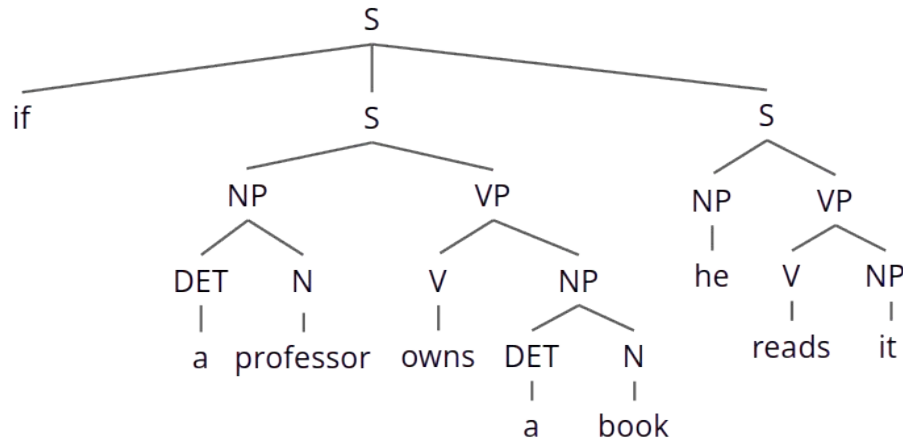
Construction rules: conditionals

- Given a reducible condition α in DRS K , with $[S \text{ if } [S \beta] \text{ (then) } [S \gamma]]$ as a substructure:
 - (i) remove α from C_K
 - (ii) add $K_1 \Rightarrow K_2$ to C_K , such that:
 - $K_1 = (\emptyset, \{\beta\})$
 - $K_2 = (\emptyset, \{\gamma\})$

Remark: $K_1 \Rightarrow K_2$ is called a **duplex condition**— K_1 is the **antecedent** DRS and K_2 is the **consequent** DRS

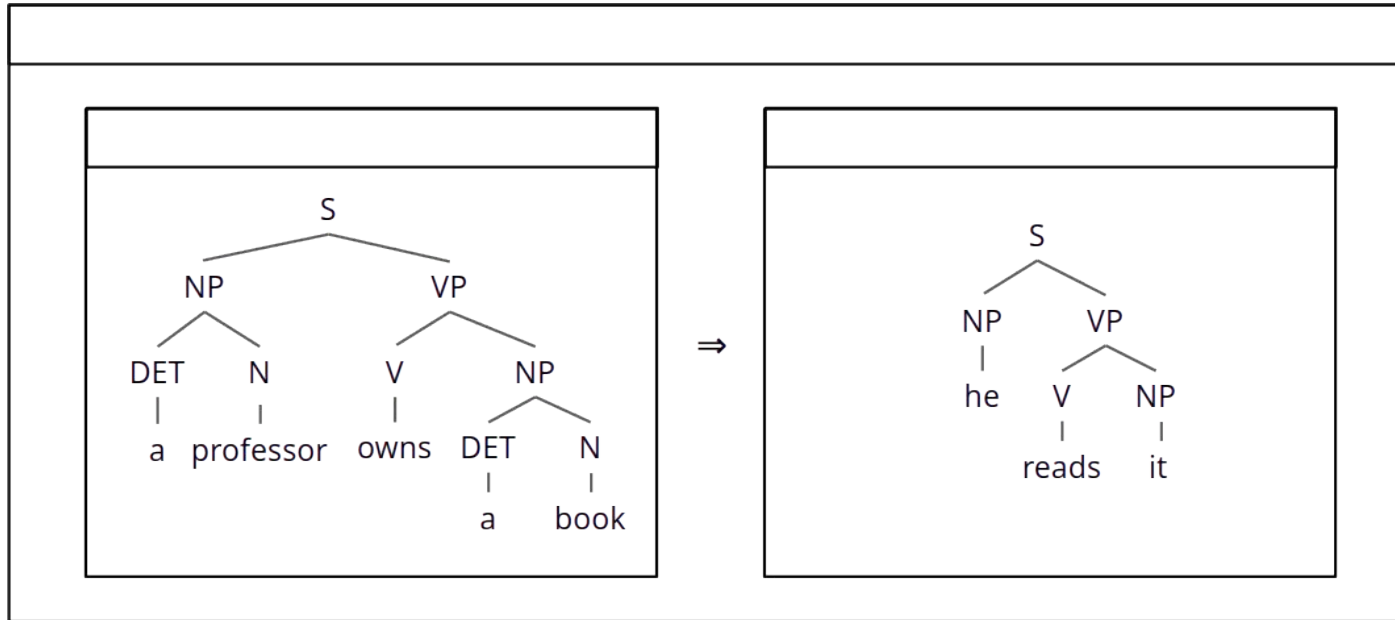
Construction rules: conditionals

“if a professor owns a book, he reads it”



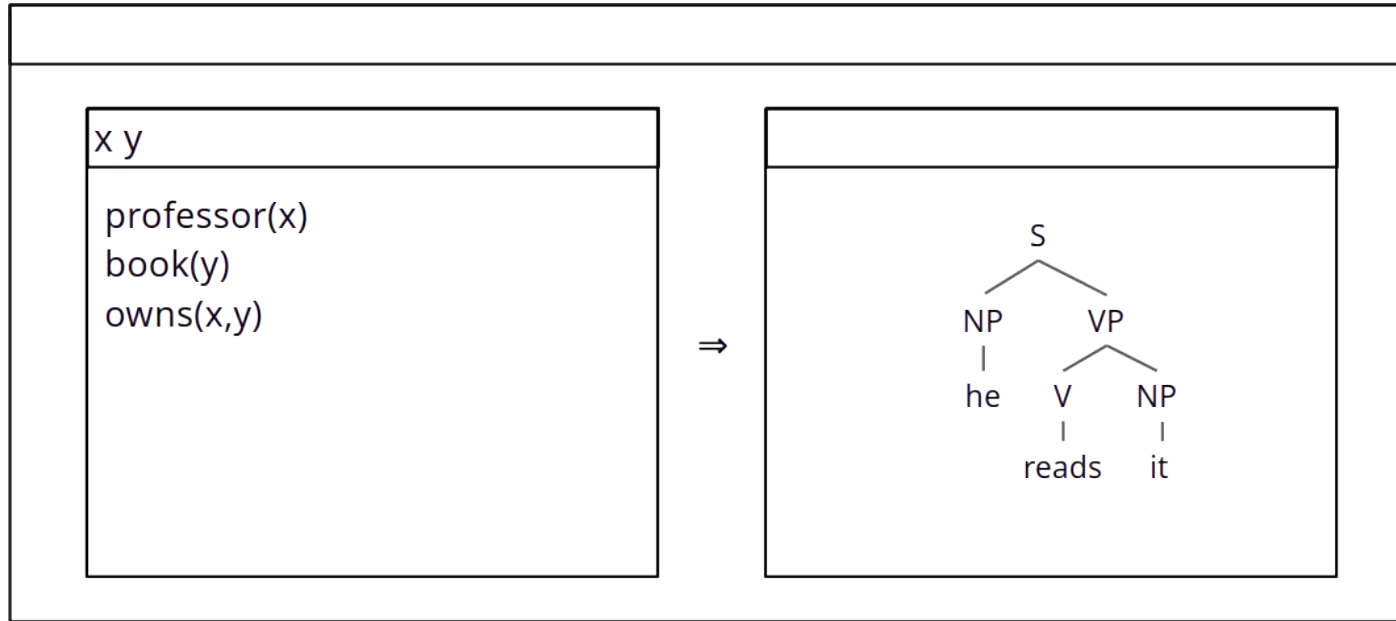
Construction rules: conditionals

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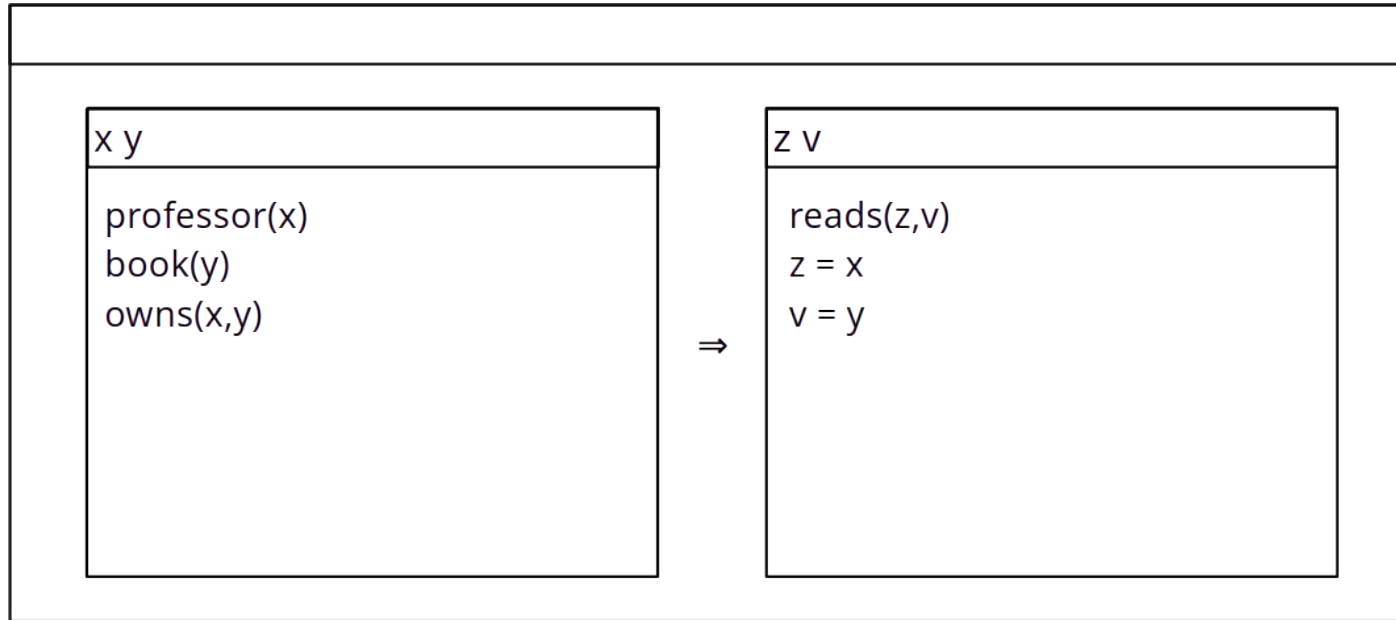
Construction rules: conditionals

“if a professor owns a book, he reads it”



Construction rules: conditionals

“if a professor owns a book, he reads it”



Construction rules: universal NPs

- given a reducible condition α in DRS K , with $[S [NP \beta] [VP \gamma]]$ or $[VP [V \gamma] [NP \beta]]$ as a substructure, such that $\beta = \varepsilon \delta$, where ε is a universal quantifier:
 - (i) remove α from C_K
 - (ii) add $K_1 \Rightarrow K_2$ to C_K , such that:
 - $K_1 = (\{x\}, \{\delta(x)\})$
 - $K_2 = (\emptyset, \{\alpha'\})$
 - where α' is obtained from α by replacing β with x

Construction rules: negation

- given a reducible condition α in DRS K , with $[S \beta [VP \text{ doesn't } [VP \gamma]]]$ as a substructure
 - (i) remove α from C_K
 - (ii) add $\neg K_1$ to C_K , such that:
 - $K_1 = (\emptyset, \{[S \beta [VP \gamma]]\})$

Construction rules: clausal disjunction

- given a reducible condition α in DRS K , with $[[S \ \beta]$ or $[S \ \gamma]]$ as a substructure
 - (i) remove α from C_K
 - (ii) add $K_1 \vee K_2$ to C_K , such that:
 - $K_1 = (\emptyset, \{\beta\})$
 - $K_2 = (\emptyset, \{\gamma\})$

From text to DRS

Text

$\Sigma = \langle S_1, S_2, \dots, S_n \rangle$

Syntactic analysis

$\downarrow \quad \downarrow \quad \quad \downarrow$
 $P(S_1) \quad P(S_2) \quad \dots \quad P(S_n)$

DRS

$\downarrow \quad \downarrow \quad \quad \downarrow$
 $K_1 \longrightarrow K_2 \longrightarrow \dots \longrightarrow K_n \longrightarrow \text{Interpretation by model embedding: truth conditions of } \Sigma$

DRS interpretation: model embedding

- Given a DRS $K = (U_K, C_K)$, with $U_K \subseteq U_D$:
 - Let $M = (U_M, V_M)$ be a FOL model structure that is **appropriate for K** , i.e. a model structure that provides interpretations for all predicates and relations in K
- K is true in M iff:
 - there exists an **embedding function** for K in M that verifies all conditions in K
- An embedding function for DRS K in model M is defined as:
 - a (partial) function $f: U_D \rightarrow U_M$ such that $U_K \subseteq \text{dom}(f)$

Verifying by embedding

$$g \supseteq_U f := \\ (dom(g) = dom(f) \cup U) \wedge \\ \forall x[x \in dom(f) \rightarrow f(x) = g(x)]$$

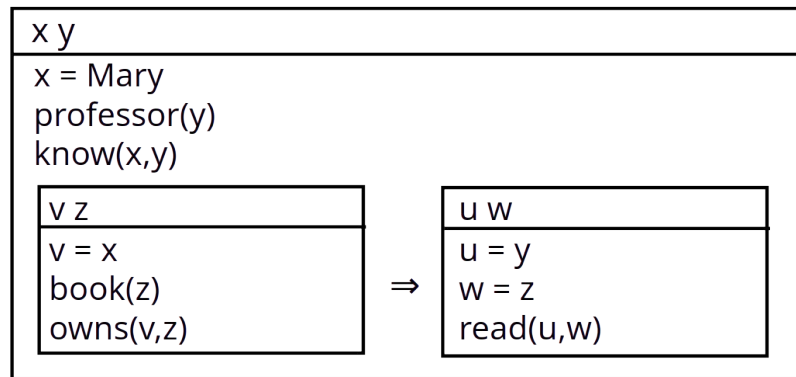
- An embedding f of K in M **verifies** K in M ($f \models_M K$) iff f verifies every condition $\alpha \in C_K$ ($f \models_M K$ for all $\alpha \in C_K$)
 - $f \models_M R(x_1, \dots, x_n)$ iff $(f(x_1), \dots, f(x_n)) \in V_M(R)$
 - $f \models_M x = y$ iff $f(x) = f(y)$
 - $f \models_M x = a$ iff $f(x) = V_M(a)$
 - $f \models_M \neg K_1$ iff there is no $g \supseteq_{U_{K_1}} f$ such that $g \models_M K_1$
 - $f \models_M K_1 \vee K_2$ iff there is a $g_1 \supseteq_{U_{K_1}} f$ such that $g_1 \models_M K_1$ or there is a $g_2 \supseteq_{U_{K_2}} f$ such that $g_2 \models_M K_2$
 - $f \models_M K_1 \Rightarrow K_2$ iff for all $g_1 \supseteq_{U_{K_1}} f$ such that $g_1 \models_M K_1$: there is a $g_2 \supseteq_{U_{K_2}} g_1$ such that $g_2 \models_M K_2$

Verifying by embedding: example

- “Mary knows a professor. If she owns a book, he reads it.” is true in

$M = (U_M, V_M)$ iff there is an $f: U_D \rightarrow U_M$,
(with $\{x, y\} \subseteq \text{dom}(f)$) s.t.:

- $f(x) = V_M(\text{mary})$ and $f(y) \in V_M(\text{professor})$
- and $(f(x), f(y)) \in V_M(\text{know})$
- and for all $g \supseteq_{\{v, z\}} f$ such that $g(v) = g(x)$, $g(z) \in V_M(\text{book})$ and $(g(v), g(z)) \in V_M(\text{own})$:
 - there exists $h \supseteq_{\{u, w\}} g$ s.t. $h(u) = h(y)$, $h(w) = h(z)$, and $(h(u), h(w)) \in V_M(\text{read})$



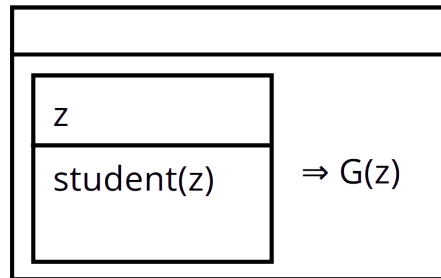
DRT and compositionality

- DRT is a **representational** theory of meaning
 - Structural information that cannot be reduced to truth conditions is required to compute the semantic value of discourses
- DRT is **non-compositional** on truth conditions (in the traditional sense)
 - The difference in discourse-semantic status of the text pairs is not predictable through the truth conditions of its component sentences
- But wait a minute... can't we just combine type theoretic semantics and DRT?
 - Use λ -abstraction and reduction as before, but where the target (type t) representations are DRSs, not formulas from type theory (or FOL)
 - This is called **λ -DRT**

λ -DRT

- An expression in λ -DRT consists of a lambda prefix and a partially instantiated DRS

- “every student” :: $\langle \langle e, t \rangle, t \rangle \mapsto \lambda G.$



- Alternative notation: $\lambda G.[\emptyset \mid [z \mid student(z)] \Rightarrow G(z)]$
 - “works” :: $\langle e, t \rangle \mapsto \lambda x.[\emptyset \mid work(x)]$

λ -DRT: β -reduction

- “every student works”

$$\mapsto \lambda G.[\emptyset \mid [z \mid \textit{student}(z)] \Rightarrow G(z)] (\lambda x.[\emptyset \mid \textit{work}(x)])$$
$$\Rightarrow_{\beta} [\emptyset \mid [z \mid \textit{student}(z)] \Rightarrow (\lambda x.[\emptyset \mid \textit{work}(x)])(z)]$$
$$\Rightarrow_{\beta} [\emptyset \mid [z \mid \textit{student}(z)] \Rightarrow [\emptyset \mid \textit{work}(z)]]$$

- **Question:** how do we define conjunction on DRSs?

Simple DRS merge: first try

- The **merge** operation (notation: $K_1 + K_2$) on two DRSs combines the universes and conditions of both DRSs into a new DRS
- Let $K_1 = [U_1 \mid C_1]$ and $K_2 = [U_2 \mid C_2]$:
 - $K_1 + K_2 = [U_1 \cup U_2 \mid C_1 \cup C_2]$

Compositional analysis with merge

- “a student” $\mapsto \lambda G.([z \mid \text{student}(z)] + G(z))$
- “works” $\mapsto \lambda x.[\emptyset \mid \text{work}(x)]$
- “a student works” $\mapsto \lambda G.([z \mid \text{student}(z)] + G(z))(\lambda x.[\emptyset \mid \text{work}(x)])$
 - $\Rightarrow_{\beta} [z \mid \text{student}(z)] + \lambda x.[\emptyset \mid \text{work}(x)](z)$
 - $\Rightarrow_{\beta} [z \mid \text{student}(z)] + [\emptyset \mid \text{work}(z)]$
 - $\Rightarrow_{\beta} [z \mid \text{student}(z), \text{work}(z)]$

Compositional analysis with merge

- “Mary” $\mapsto \lambda G.([z \mid z = \text{mary}] + G(z))$
- “she” $\mapsto \lambda G.([v \mid v = z] + G(v))$
- “Mary works. She is successful.”
 - $\mapsto \lambda K \lambda K' (K + K')([z \mid z = \text{mary}, \text{work}(z)])([v \mid v = z, \text{successful}(v)])$
 - $\Rightarrow_{\beta} \lambda K' ([z \mid z = \text{mary}, \text{work}(z)] + K')([v \mid v = z, \text{successful}(v)])$
 - $\Rightarrow_{\beta} [z \mid z = \text{mary}, \text{work}(z)] + [v \mid v = z, \text{successful}(v)]$
 - $\Rightarrow [z \ v \mid z = \text{mary}, \text{work}(z), v = z, \text{successful}(v)]$

DRS merge: second try (directional)

- The merge operation on two DRSs combines the universes and conditions of both DRSs into a new DRS
- Let $K_1 = [U_1 \mid C_1]$ and $K_2 = [U_2 \mid C_2]$:
 - $K_1 + K_2 = [U_1 \cup U_2 \mid C'_1 \cup C_2]$
 - where: C'_1 is C_1 such that all free variables in the conditions $\gamma \in C_1$ that also occur as discourse referents $u \in U_2$ are α -converted to new variables
- under this definition, merge is directional:

$$K_1 + K_2 \not\equiv K_2 + K_1$$

Variable capturing

- In λ -DRT, discourse referents are captured via the interaction of β -reduction and DRS-binding:

$$\begin{aligned} & \lambda K'([z \mid \text{student}(z), \text{work}(z)] + K')([v \mid v = z, \text{successful}(v)]) \\ & \Rightarrow_{\beta} [z \mid \text{student}(z), \text{work}(z)] + [v \mid v = z, \text{successful}(v)] \\ & \Rightarrow_{\beta} [z \ v \mid \text{student}(z), \text{work}(z), v = z, \text{successful}(v)] \end{aligned}$$

- But the β -reduced DRS must be equivalent to the original DRS!
 - This means that the potential for capturing discourse referents must be captured in the interpretation of λ -DRSs
 - Possible, but tricky